

Tensor networks for neuro-symbolic AI

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joint work with Janina Schütte, Maximilian Fröhlich and Martin Eigel

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Weierstraß Institute for Applied Analysis and Stochastics (WIAS)

NESYRE Workshop

September 15, 2026

- ▶ Tensor networks as a unifying framework for AI

- ▶ Motivation from maximum entropy principles

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- ▶ Motivation from maximum entropy principles

Motivation: Automation of accounting at DATEV

Accounts need to be selected for invoices¹:

Features
extracted by OCR

Invoice
to be booked

Generated
accounting proposal

The screenshot displays the DATEV software interface. At the top, there's a header bar with various tabs like 'BL', 'WKZ', 'Umsatz', 'S', 'GU', 'BU', 'Gegenkonto', 'Belegfeld 1', 'Belegfeld 2', 'Datum', 'Konto', 'KOST1', 'KOST2', and 'KOST-Menge'. Below this, a window titled 'Digitalen Beleg buchen (Rechnungen Februar / Februar / 3)' is open. It shows an OCR-extracted invoice from 'Elektro Kühn & Sohn' dated 03.02.2024. The invoice details include a net amount of 642.80 and a total amount of 735.5084. A 'MUSTER' watermark is visible across the invoice. Below the invoice, a table lists items: '100-Reparatur' (100,00 €) and 'Umlauf vor Ort' (60,00 €). A 'Rechnungswert (brutto)' of 160,00 € is shown, along with a 'Zugl. 19 % MwSt' of 302,88 €. The total 'Rechnungswert (brutto)' is 462,88 €. At the bottom, a 'Generated accounting proposal' is shown, featuring a table with columns for 'GU', 'WKZ', 'Umsatz', 'BU', 'Gegenkonto', 'H', 'Belegfeld 1', 'Datum', 'Konto', 'KOST1', 'KOST2', and 'Skonto'. The proposal includes a 'Buchungstext' of 'Elektro Kühn & Sohn 02/24' and a 'Buchungssatz' of 'Elektro Kühn & Sohn'. The 'Konto' is '6430 Sonstige Reparaturen u. Instandhaltungen' and the 'Skonto' is '1.053.15 H'.

¹ <https://www.datev.de/web/de/steuerberatung/loesungen/rechnungswesen/buchfuehrung-erstellen/automatisierungsservices-rechnungswesen/>

Toy accounting system

Assume you want to book one of two accounts A1 and A2 depending on a feature F of an invoice, based on two rules:

- ▶ Exactly one account must be booked.
 - ▶ If feature F is present on the invoice, the account A1 is typically booked.
-
- ▶ Accounting rules are formalised in propositional logics, triggered by receipts properties.
 - ▶ These rules can either be hard restrictions, or prone to exceptions.

We propose tensor (networks) to represent hard and soft accounting rules in a unified framework.

Factored representations lead to tensors

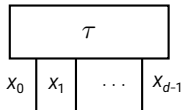
Factored representations contain variables $X_{[d]} = (X_0, \dots, X_{d-1})$, each with a finite set of states $[m_k] = \{0, \dots, m_k - 1\}$. The set of states is then the cartesian product

$$\bigtimes_{k \in [d]} [m_k] = \{0, \dots, m_0\} \times \dots \times \{0, \dots, m_{d-1}\}.$$

Information about the system (e.g. possibility or uncertainty of a state) is stored in a **tensor**

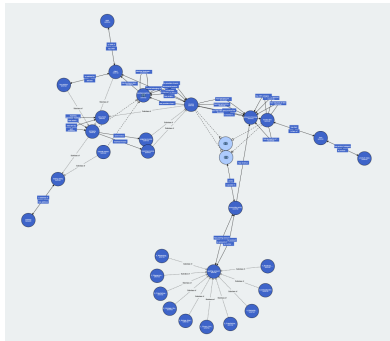
$$\tau [X_{[d]}] : \bigtimes_{k \in [d]} [m_k] \rightarrow \mathbb{R}$$

where we denote its coordinates by $\tau [X_{[d]}] (x_{[d]}) = \tau [X_{[d]} = x_{[d]}]$. We represent tensors by diagrams with legs decorated by the variables:

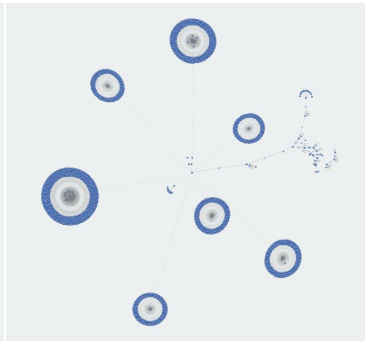


Outlook: Productive accounting ontology

The variables are described by the DATEV accounting ontology²:



Coarse Visualization



Detailed Visualization

The curse of dimensionality prevents naive tensor representations.

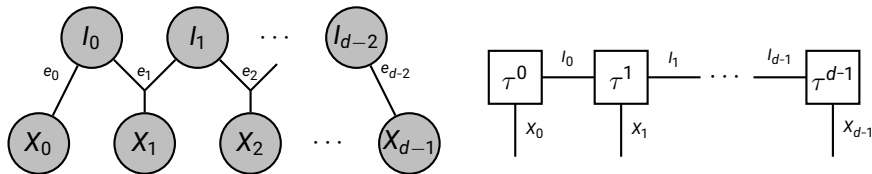
²<https://github.com/EnexaProject/enexa-use-case-1-business-software-solutions>

Tensor network decompositions

To overcome the curse of dimensionality, tensors $\tau^{\mathcal{G}} [X_{\mathcal{V}}]$ are decomposed into core tensors $\tau^e [X_e]$. Tensor networks on hypergraphs $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ are factorizations

$$\tau^{\mathcal{G}} [X_{\mathcal{V}} = \mathbf{x}_{\mathcal{V}}] = \prod_{e \in \mathcal{E}} \tau^e [X_e = \mathbf{x}_e] .$$

Example: Tensor-Train (TT) format with hidden variables $l_{[d-1]}$ (to be summed over).



Beyond black-boxes: Hypergraphs encode symbolic structure.

- Probabilistic perspective: [Conditional independencies](#).
- Neural and logical perspective: [Function decompositions](#).

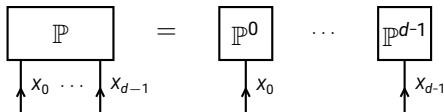
Probability distributions

Joint distributions of variables are tensors $\mathbb{P} [X_{[d]}]$ with non-negative coordinates $\mathbb{P} [X_{[d]} = x_{[d]}]$ (the probabilities of states $x_{[d]}$) such that

$$\sum_{x_{[d]} \in \times_{k \in [d]} [m_k]} \mathbb{P} [X_{[d]} = x_{[d]}] = 1.$$

Example: If the variables are independent on each other we have a decomposition by an elementary tensor network

$$\mathbb{P} [X_{[d]} = x_{[d]}] = \prod_{k \in [d]} \mathbb{P}^k [X_k = x_k]$$



The curse of dimensions is mitigated by independence!

Pairwise independence is a very restrictive assumption:

- ▶ AI requires dependencies for non-trivial inference.
- ▶ Usually variables depend on each other.

Conditional independence models variable dependence conditioned on states of a third variable.

Graphical models factorize distributions into tensors $\tau^e [X_e]$, coordinate are then retrieved by

$$\mathbb{P} [X_{\mathcal{V}} = x_{\mathcal{V}}] = \prod_{e \in \mathcal{E}} \tau^e [X_e = x_e] .$$

Hammersley-Clifford Theorem (informal)

Node separation in hypergraphs correspond with conditional independence in the distribution.

Example: Recommending a student

Consider a distribution of a student's intelligence X_I , SAT score X_S , test difficulty X_D , grade X_G and recommendation X_L .

The corresponding hypergraph is:

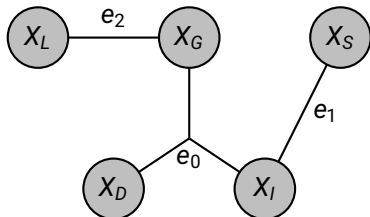
Let us assume the independencies:

- "The SAT score depends only on the student's intelligence":

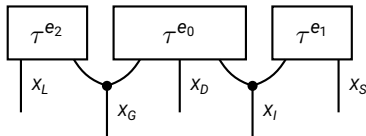
$$(X_S \perp X_{\{D,G,L\}}) \mid X_I$$

- "The recommendation depends only on the grade":

$$(X_L \perp X_{\{D,I,S\}}) \mid X_G$$



$\mathbb{P}[X_L, X_G, X_D, X_I, X_S]$ is decomposed into:



Conditional independencies as tensor decompositions

When we condition on the grade x_G , the distribution is:

$$\frac{1}{\mathbb{P}[X_G=x_G]} \cdot \mathbb{P}[X_L, X_D, X_I, X_S | X_G = x_G] =$$

Conditional independencies as tensor decompositions

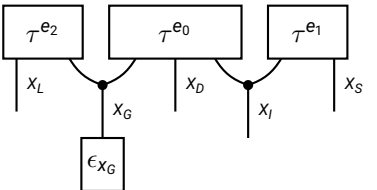
When we condition on the grade x_G , the distribution is:

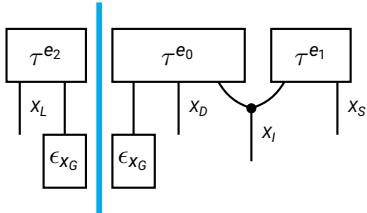
$$\frac{1}{\mathbb{P}[X_G = x_G]} \cdot \mathbb{P}[X_L, X_D, X_I, X_S | X_G = x_G] =$$

$$=$$

Conditional independencies as tensor decompositions

When we condition on the grade x_G , the distribution is:

$$\frac{1}{\mathbb{P}[X_G=x_G]} \cdot \mathbb{P}[X_L, X_D, X_I, X_S | X_G = x_G] =$$


$$=$$


Interpretation:

The variable X_L is independent of X_D, X_I, X_S conditioned on X_G .

Example: Integer summation in m -adic representation

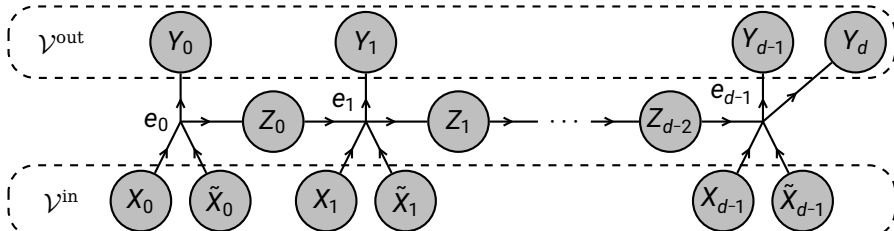
Any number $n \in [m^d]$ has a unique m -adic representation $x_{[d]}$ such that

$$n = \sum_{k \in [d]} x_k \cdot m^k.$$

Given two numbers $x_{[d]}$ and $\tilde{x}_{[d]}$ we want the tuple $y_{[d+1]}$ such that

$$\sum_{k \in [d+1]} y_k \cdot m^k = \sum_{k \in [d]} (x_k + \tilde{x}_k) \cdot m^k.$$

The function decomposes into half-adders with carry bit variables $Z_{[d-1]}$:



Aim: Find tensor network decompositions based on circuit graphs.

We define the **basis encoding** of a function

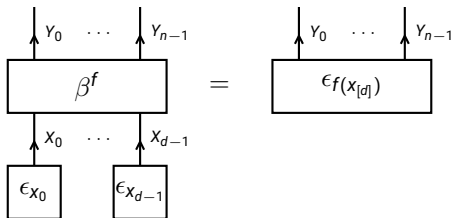
$$f : \prod_{k \in [d]} [m_k] \rightarrow \prod_{\ell \in [p]} [m_\ell]$$

as the sum over one-hot encodings

$$\beta^f [Y_{[p]}, X_{[d]}] = \sum_{x_{[d]} \in \prod_{k \in [d]} [m_k]} \epsilon_{f(x_{[d]})} [Y_{[p]}] \otimes \epsilon_{x_{[d]}} [X_{[d]}] .$$

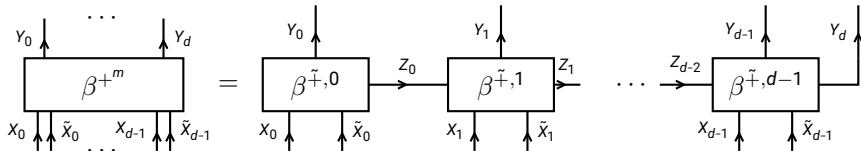
Basis calculus: Contraction with encoded input yields the encoded output.

$$\sum_{x_{[d]} \in \prod_{k \in [d]} [m_k]} \beta^f [Y_{[p]}, X_{[d]} = x_{[d]}] \cdot \epsilon_{x_{[d]}} [X_{[d]} = x_{[d]}] = \epsilon_{f(x_{[d]})} [Y_{[p]}]$$

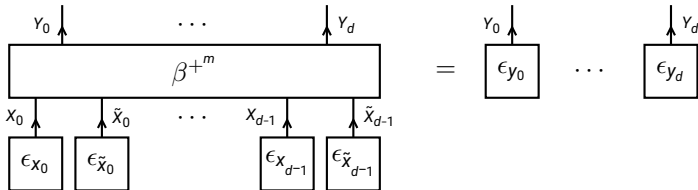


Decomposition of basis encodings

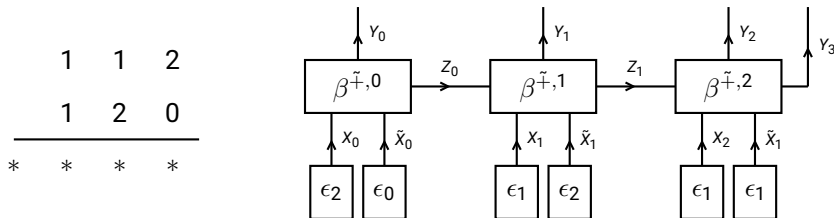
Oriented on the circuit of half-adders we find a tensor network decomposition:



Basis Calculus: Contraction with one-hot encoded input integers yields the one-hot encoded sum:



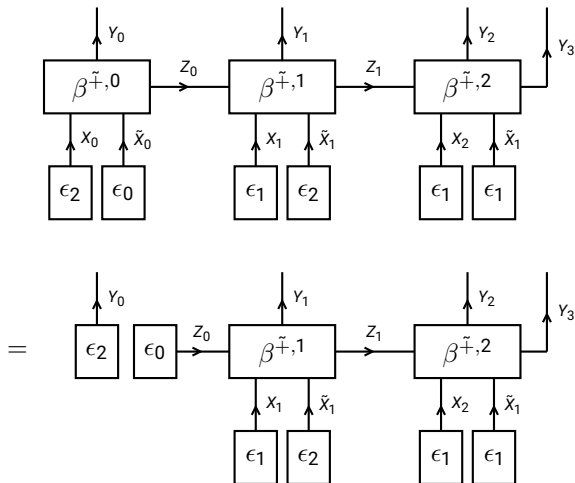
As an example, let us consider the sum of $x_{[3]} = (2, 1, 1)$ ($n = 14$) and $\tilde{x}_{[3]} = (0, 2, 1)$ ($n = 15$) in basis $m = 3$:



Efficient basis calculus

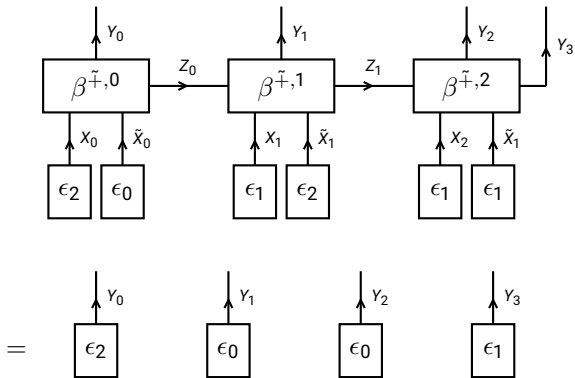
As an example, let us consider the sum of $x_{[3]} = (2, 1, 1)$ ($n = 14$) and $\tilde{x}_{[3]} = (0, 2, 1)$ ($n = 15$) in basis $m = 3$:

$$\begin{array}{rrrr}
 1 & 1 & 2 & \\
 1 & 2_0 & 0 & \\
 \hline
 * & * & * & 2
 \end{array}$$



As an example, let us consider the sum of $x_{[3]} = (2, 1, 1)$ ($n = 14$) and $\tilde{x}_{[3]} = (0, 2, 1)$ ($n = 15$) in basis $m = 3$:

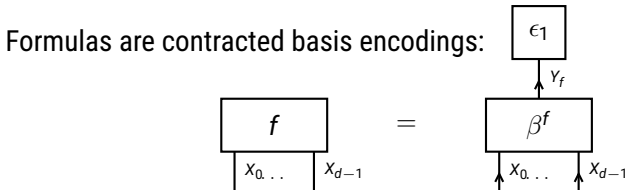
$$\begin{array}{r}
 \begin{array}{cccc}
 & 1 & 1 & 2 \\
 & 1_1 & 2_0 & 0 \\
 \hline
 1 & 0 & 0 & 2
 \end{array}
 \end{array}$$



Decomposition of propositional formulas

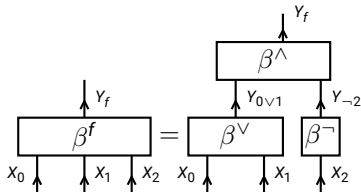
Propositional formulas are tensors $f [X_{[d]}]$ with coordinates

$$f [X_{[d]} = x_{[d]}] = \begin{cases} 1 & \text{if } x_{[d]} \text{ is a model of } f \\ 0 & \text{else.} \end{cases}$$



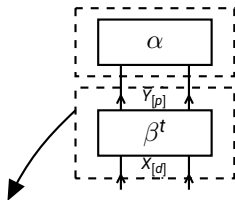
The basis encodings decompose into networks of connectives.

Example: The basis encoding of $f [X_{[3]}] = (X_0 \vee X_1) \wedge \neg X_2$ is decomposed as:



Computation-Activation Networks (CompActNets)

A novel tensor network architecture captures logical and probabilistic models³:



Activation of a statistic by an activation core, for **knowledge bases**

$$\alpha [Y_{[p]}] = \bigotimes_{\ell \in [p]} \begin{pmatrix} 0 \\ 1 \end{pmatrix} [Y_{\ell}]$$

and for **exponential families**

$$\alpha [Y_{[p]}] = \bigotimes_{\ell \in [p]} \begin{pmatrix} 1 \\ \exp [\theta [L = \ell]] \end{pmatrix} [Y_{\ell}]$$

Computation of a statistic

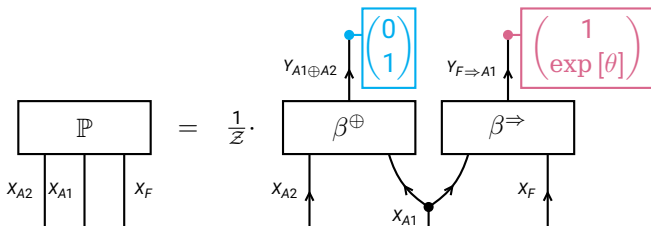
$$t : \bigtimes_{k \in [d]} [m_k] \rightarrow \bigtimes_{\ell \in [p]} [p_{\ell}]$$

by basis encoding

³Goessmann, Schütte, Fröhlich and Eigel: A tensor network formalism for neuro-symbolic AI, arXiv:2601.15442

Toy accounting system as a CompActNet

We represent the toy accounting system as a Computation-Activation Network:



- ▶ The hard activation vector $\begin{pmatrix} 0 \\ 1 \end{pmatrix} [y_{A1 \oplus A2}]$ ensures that the first rule is always true.
- ▶ The soft activation vector $\begin{pmatrix} 1 \\ \exp[\theta] \end{pmatrix} [y_{F \Rightarrow A1}]$ introduces the trainable parameter $\theta \in \mathbb{R}$ controlling the satisfaction probability of the second rule.

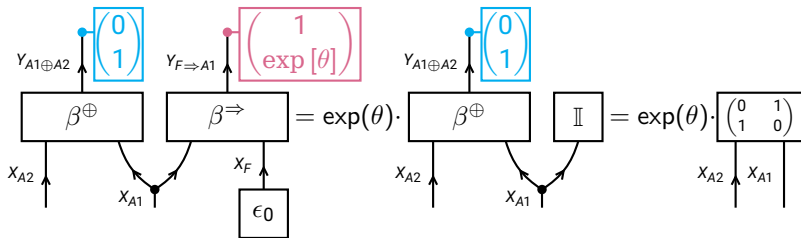
Inference by contractions

Contractions of tensor networks $\{\tau^e[X_e] : e \in \mathcal{E}\}$ keeping $\mathcal{U} \subset \mathcal{V}$ open are tensors $\langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[X_{\mathcal{U}}]}$ with coordinates

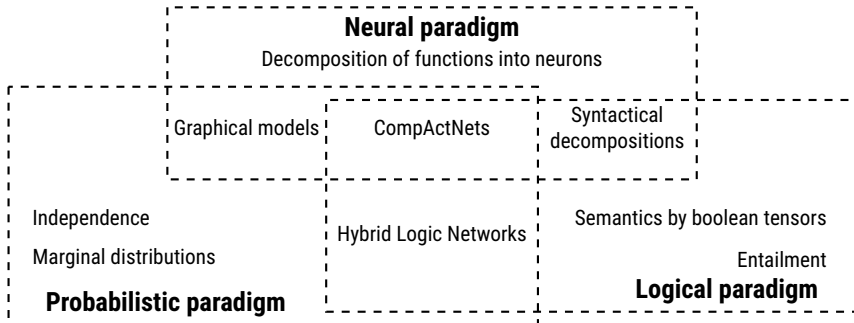
$$\langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[X_{\mathcal{U}}=x_{\mathcal{U}}]} = \sum_{x_{\mathcal{V}/\mathcal{U}}} \prod_{e \in \mathcal{E}} \tau^e[X_e = x_e] .$$

- **Probabilistic inference:** Contractions are marginal probabilities.
- **Logical inference:** Contractions are model counts, which decide entailment.

Example: Distribution of X_{A1} and X_{A2} conditioned on $X_F = 0$ is parallel to



Tensor networks capture the different paradigms of AI⁴:



⁴Goessmann, Schütte, Fröhlich and Eigel: A tensor network formalism for neuro-symbolic AI, arXiv:2601.15442

When turning from **factored representations** (e.g. propositional logics) to **structured representations** (e.g. first-order logics) challenges arise:

- ▶ States of a system contain (varying numbers of) objects
- ▶ Formulas contain variables and quantifiers

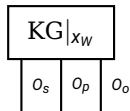
To describe probability distributions over such worlds we encounter:

- ▶ **Substitution structure**: States itself are described by tensors, using term variables O
- ▶ **Semantic structure**: Variables X enumerate all possible states

Which tricks are used to reason in such large spaces?

Knowledge graphs can be represented by order-3 tensors, with variables O_s, O_p, O_o enumerating the nodes:

$$\text{KG}|_{x_W} [O_s = o_s, O_p = o_p, O_o = o_o] \\ = \begin{cases} 1 & \text{if the triple } (o_s, o_p, o_o) \text{ exists} \\ 0 & \text{else} \end{cases} .$$



The tensor structure has been exploited to

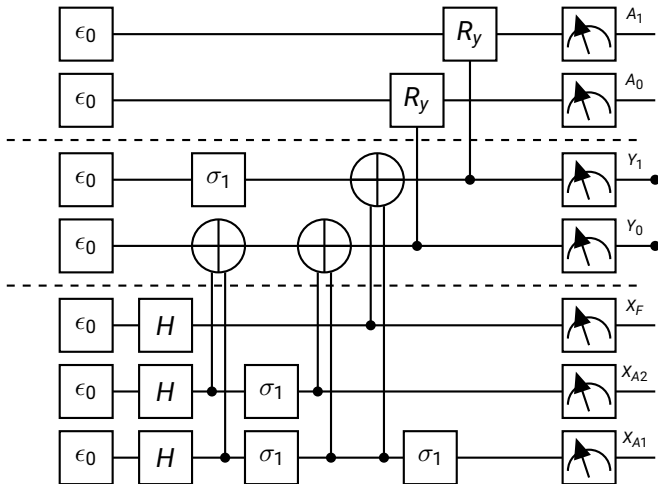
- ▶ Train embedding models (RESICAL, ComplEx, DistMult)
- ▶ Answer SPARQL queries efficiently (tentriss)
- ▶ Derive statistical models⁵

⁵ Nickel et al: A review of relational machine learning for knowledge graphs. Proceedings of the IEEE 104(1):11-33

Outlook: Quantum inference for CompActNets

Quantum computers are naturally tensor network contractors.

Example: Quantum rejection sampling of the toy accounting system by



- ▶ Tensor networks as a unifying framework for AI
- ▶ Motivation from maximum entropy principles

Toy accounting system

Assume you want to book one of two accounts A1 and A2 depending on a feature F of an invoice, based on two rules:

- ▶ Exactly one account must be booked.
- ▶ If feature F is present on the invoice, the account A1 is typically booked.

Lets assume the following satisfaction rates of the toy accounting system on an invoice dataset:

Always exactly one account has been booked: $\mu[L = 0] = 1$

Feature F implies in most cases account A1: $\mu[L = 1] = 0.9$

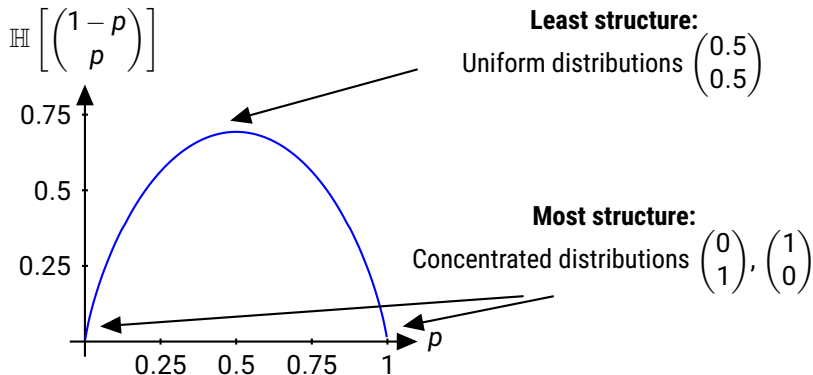
What is the "simplest" distribution reproducing these satisfaction rates?

Entropy: Quantifying the structure of a distribution

The entropy of a distribution $\mathbb{P} [X_{[d]}]$ is a measure of its structure:

$$\mathbb{H} [\mathbb{P} [X_{[d]}]] = - \sum_{x_{[d]} \in \times_{k \in [d]} [m_k]} \mathbb{P} [X_{[d]} = x_{[d]}] \cdot \ln [\mathbb{P} [X_{[d]} = x_{[d]}]]$$

Example: Entropy of Bernoulli distributions



Maximum entropy principle

Among those distributions reproducing certain characteristics of observed data, choose the one with the least structure (=maximum entropy).

To simplify notation we introduce selection encodings of functions t as tensors with coordinates

$$\sigma^t [X_{[d]} = x_{[d]}, L = \ell] = t_\ell(x_{[d]}).$$

Formally the maximum entropy problem is

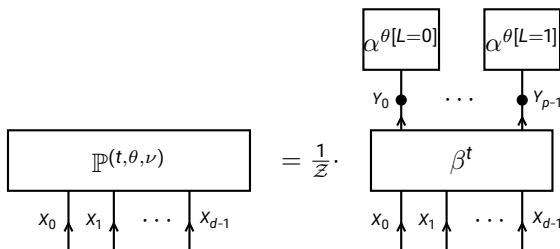
$$\operatorname{argmax}_{\mathbb{P}[X_{[d]}]} \mathbb{H} [\mathbb{P} [X_{[d]}]] \quad \text{s.t.} \quad \langle \mathbb{P} [X_{[d]}], \sigma^t [X_{[d]}, L] \rangle_{[L]} = \mu [L].$$

Exponential families are maximum entropy distributions

The **exponential family** of a statistic t is the set of distributions

$$\mathbb{P}^{(t, \theta)} [X_{[d]}] = \frac{1}{\mathcal{Z}(\theta)} \exp \left[\langle \theta [L], \sigma^t [X_{[d]}, L] \rangle_{[L]} \right]$$

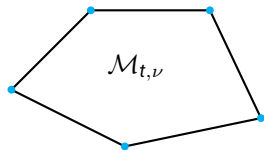
where $\theta [L] \in \mathbb{R}^p$. We find elementary activation cores $\alpha^{\theta[L=\ell]} [Y_\ell]$ to represent the members as CompActNets:



Classical result: Any member of the exponential family is a solution of the maximum entropy problem to its mean parameter $\mu [L]$.

Realizable mean parameters to a statistic t form the **mean polytope**

$$\mathcal{M}_t = \text{conv} \left(\sigma^t [X_{[d]} = x_{[d]}, L] : x_{[d]} \in \prod_{k \in [d]} [m_k] \right)$$



Exponential families only realize the relative interior of the mean polytope.

What are the solutions to the mean parameters on the boundary of the mean polytope?

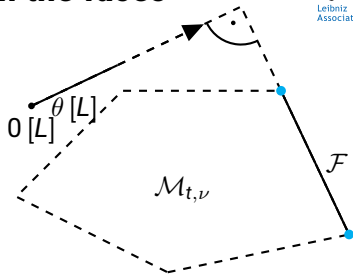
Mean parameters on the faces

Faces of the mean polytope are

$$\mathcal{F} = \operatorname{argmax}_{v \in \mathcal{M}_t} \langle \theta[L], v[L] \rangle_{[\emptyset]} \quad \mathbb{R}^p$$

and represented by cores

$$\kappa^{\mathcal{F}} [Y_{[p]}] = \sum_{v \in \mathcal{F} \cap X_{\ell \in [p]}[n_\ell]} \epsilon_v [Y_{[p]}] .$$



Theorem (Generic maximum entropy representation⁶)

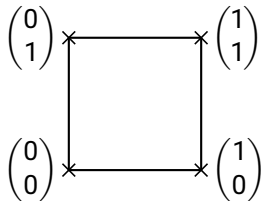
For any mean parameter on the relative interior of a face $\mathcal{F}$, we find a parameter $\theta[L]$ such that the corresponding maximum entropy is

$$\frac{1}{Z} \cdot \left\langle \beta^t [Y_{[p]}, X_{[d]}] , \kappa^{\mathcal{F}} [Y_{[p]}] , \alpha^\theta [Y_{[p]}] \right\rangle_{[X_{[d]}]} .$$

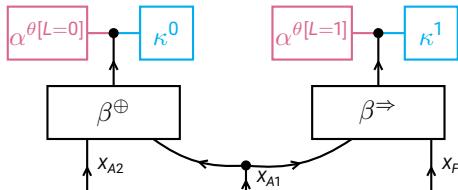
⁶Goessmann and Eigel: Tensor network repr. of discrete maximum entropy distr. via mean polytopes, arXiv:2609.07184

Toy accounting example

The mean polytope is

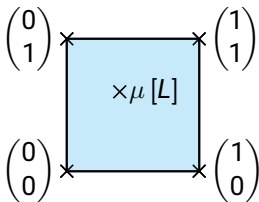


Any solution of the maximum entropy problem is a CompActNet with elementary activation:

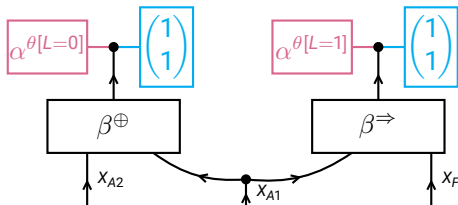


Toy accounting example

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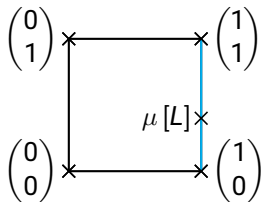


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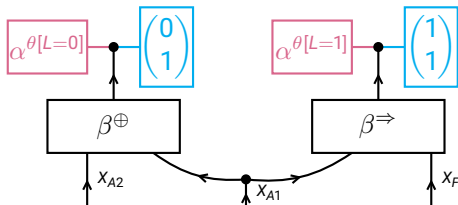


Toy accounting example

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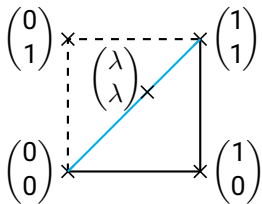
Any solution of the maximum entropy problem is a CompActNet with elementary activation:



Consider the statistic

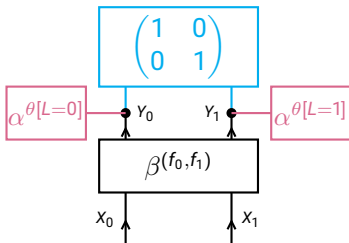
$$f_0[X_0, X_1] = (X_0 \wedge X_1) \quad , \quad f_1[X_0, X_1] = (X_0 \Rightarrow X_1)$$

The mean polytope of this statistic is



The blue face cannot be represented by an elementary tensor.

Representation of the maximum entropy distribution for a mean parameter in the blue face:



Outlook: Application of the maximum entropy approach

Maximum likelihood estimation:

- ▶ Numerical instabilities arise when the mean is not on the parametrised face
- ▶ Choosing a sufficient tensor format guarantees a solution

Probabilistic success guarantees:

- ▶ Hard constraints tend to be overestimated
- ▶ Derive uniform concentration bounds for $\mu [L]$ to guarantee correct estimation⁷

⁷ Goessmann: Tensor networks for Efficient and Explainable AI, Technical report on the ENEXA project, <https://github.com/EnexaProject/enexa-tensor-reasoning-documentation/>

Further material available at github.com/tnreason:

- ▶ Python library `tnreason`: Reasoning with tensor networks, also available on <https://pypi.org/project/tnreason/>
- ▶ Course on "Foundations of Neuro-Symbolic AI" (Summer term 2026, THWS): Introduction to symbolic and neural reasoning based on tensor networks

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- ▶ **ENEXA** (2023-2025): Efficient and Explainable Learning on Knowledge Graphs, funded by Horizon 2020 program
- ▶ **QROM** (2025-2028): Quantum Read-Once-Memory, funded by BMBF